Recap:

$$a_n + c_{n-1} a_{n-1} = Kr^n$$

general sol^{n} of the hom. part

particular sol^{n}

general sol^{n} for the nonhom recurrence.

K is constant, $n \in \mathbb{Z}^+$

$a_n + c_{n-1} a_{n-1} = 0$

If r^n is not a sol^{n} of this hom part then

$a_n^{(p)} = Ar^n$ $A \leftarrow$ constant.

However if r^n is a sol^{n} of the hom. gen part $\rightarrow$ $a_n^{(p)} = Bnr^n$, $B \leftarrow$ constant

Linear dependence with the sol^{n} of the hom. part.

Second order:

$$a_n + C_{n-1} a_{n-1} + C_{n-2} a_{n-2} = k r^n$$

k is a constant & $n \in Z^+$

a) $a_n^{(p)} = A r^n$, A is constant, when r^n is not itself a soln of the hom. part.

b) $a_n^{(p)} = Bn r^n$, B is a constant &

$a_n^{(h)} = C_1 r^n + C_2 r_1^n$, $r_1 \neq r$

c) $a_n^{(p)} = Cn^2 r^n$, C is a constant,

$a_n^{(h)} = (C_1 + C_2 n) r^n$.

E.g. 1.

$$a_{n+1} = 2a_n + 1, \quad n \geq 0, \quad a_0 = 0.$$

$$a_{n+1} - 2a_n = 1.$$

$$a_n^{(h)} = c(2^n).$$

$f(n) = 1$ ≠ linearly dependent on the hom. soln

$$a_n^{(p)} = A(1)^n.$$

$a_n = A(1)^n$

$a_{n+1} = A$

$A = 2A + 1$

$A = -1.$

Total soln: $\boxed{a_n = c(2^n) - 1}$

$a_0 = 0$, $c = 1.$

$\therefore \boxed{a_n = 2^n - 1.}$ → No. of exchanges of discs in a Tower's of Hanoi

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Eg.2:

$a_{n+2} - 4a_{n+1} + 3a_n = -200$ ✓

$n \geq 0$, $a_0 = 3000$, $a_1 = 3300.$

$\boxed{a_n^{(h)} = c_1 3^n + c_2 1^n.}$

$f(n) = -200(1^n).$

$a_n^{(p)} = Bn1^n$

$B(n+2) - 4B(n+1) + 3Bn = -200.$

$B = 100.$

$$a_n = c_1 3^n + c_2 + 100n$$

$a_0 = 3000, \quad a_4 = 3300$

$$a_n = 100\,3^n + 2900 + 100n.$$

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① Generating functions (GF).

② Use this GF to solve recurrences.

Defⁿ:

Let $a_0, a_1, a_2 \ldots$ be a sequence of real nos.

The function

$$f(x) = a_0 + a_1 x + a_2 x^2 + \ldots \ldots = \sum_{i=0}^{\infty} a_i x^i$$

is called the generating function of this sequence.

$a_0, a_1 \ldots$ co-efficients

GF is a function that generates a series whose coefficients are the entries in the sequence.

$n \in Z^+$

$f(n) \quad (1+x)^n = \binom{n}{0} + \binom{n}{1} x + \binom{n}{2} x^2 + \ldots + \binom{n}{n} x^n + 0 + 0 + 0 \ldots$

↳ is the GF of the sequence

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots \binom{n}{n}, 0, 0, 0, \ldots$$

Part A:

$$1 - x^{n+1} = (1-x)(1 + x + x^2 + \dots + x^n).$$

$$\frac{1-x^{n+1}}{(1-x)} = 1 + x + x^2 + \dots + x^n.$$

if $|x| < 1$, $x^{n+1} \longrightarrow 0$ as $n \longrightarrow \infty$

$$1 = (1-x)(1 + x + x^2 + \dots + x^n).$$

$$\Rightarrow \left(\frac{1}{1-x}\right) = 1 + x + x^2 + \dots + x^n.$$

$\frac{1}{1-x}$ is the generating function of the sequence: 1, 1, 1, 1, ...

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx}\left(1 + x + x^2 + x^3 + \cdots\right)$$

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots$$

→ is the GP for the infinite seq:

1, 2, 3, ...

$$\frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + \cdots$$

$$= 0 + 1x + 2x^2 + 3x^3 + \cdots$$

→ GF for the sequence

0, 1, 2, 3, ...

$$d\left(\frac{x}{(1-x)^2}\right) = \frac{d}{dx}\left(0 + 1x + 2x^2 + \cdots\right)$$

$$\frac{x+1}{(1-x)^3} = 1 + 2^2 x + 3^2 x^2 + \cdots$$

$\xrightarrow{\text{G.F.}}$ sequence $\rightarrow 1^2, 2^2, 3^2, 4^2 \cdots$

$$\frac{x(x+1)}{(1-x)^3} \xrightarrow{\text{G.F.}} \text{seq} \rightarrow 0^2, 1^2, 2^2, 3^2, 4^2 \cdots$$

E.g.

Find the gf of the sequence

$0, 2, 6, 12, 20, 30, 42, \cdots$

$$a_0 = 0 = 0^2 + 0$$
$$a_1 = 2 = 1^2 + 1$$
$$a_2 = 6 = 2^2 + 2$$
$$\vdots \quad \vdots$$

$$a_n = n^2 + n$$

$$\frac{n(n+1)}{(1-x)^3} + \frac{n}{(1-x)^2} = \frac{2n}{(1-x)^3}$$

$n, r \in \mathbb{Z}^+$, $n \geq r > 0$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$$

$$\binom{-n}{r} = \frac{(-n)(-n-1)(-n-2)\cdots(-n-r+1)}{r!}$$

$$= \frac{(-1)^r n(n+1)(n+2)\cdots(n+r-1)}{r!}$$

$$= (-1)^r \frac{(n+r-1)!}{(n-1)!\,r!} = (-1)^r \binom{n+r-1}{r}.$$

For even real values of n, $\binom{n}{0} = 1$.

$$(1+x)^{-n} = \binom{-n}{0} + \binom{-n}{1}x + \binom{-n}{2}x^2 + \cdots$$

$$= \sum_{r=0}^{\infty} \binom{-n}{r} x^r$$

$(1+x)^{-n}$ is the gf. of the seq

$$\binom{-n}{0}, \binom{-n}{1}, \cdots\cdots$$

$$(1+x)^{-n} = \sum_{r\geq 0} (-1)^r \binom{n+r-1}{r} x^r$$

E.g. Find the coefficient of x^5 in $(1-2x)^{-7}$

Hint: $y = -2x$.

$$(1-2x)^{-7} = (1+y)^{-7} = \sum_{r \geq 0} \binom{-7}{r} y^r$$

$$= \sum_{r \geq 0} \binom{-7}{r} (-2x)^r$$

Coefficient of x^5 $= \binom{-7}{5}(-2)^5$

$$\downarrow$$

$$(-1)^5 \binom{7+5-1}{5} (-2)^5$$

$$= 14784.$$

E.g: Find the coefficient of x^{15} in

$$f(x) = (x^2 + x^3 + x^4 + \cdots)^4$$

$$x^2 + x^3 + x^4 \cdots$$

$$= x^2(1 + x + x^2 + \cdots) = \frac{x^2}{1-x}$$

$$f(x) = \left(\frac{x^2}{1-x}\right)^4 = x^8[1-x]^{-4}$$

In order to find the coefficient of $x^{15} \longrightarrow$

Find the coefficient of x^7 in

$(1-x)^{-4} \longrightarrow (-x)^7$

$$\hookrightarrow \binom{-4}{7}(-1)^7$$

$$= (-1)^7 \binom{4+7-1}{7} \cdot (-1)^7$$

$$= 120.$$

Defn:

GF : $f(x)$, $g(x)$

Define : $h(x) = f(x)g(x)$

$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots$

$g(x) = b_0 + b_1 x + b_2 x^2 + \cdots$

$h(x) = a_0 b_0 + (a_0 b + b_0 a_1)x + (a_0 b_2 + a_1 b_1 + a_2 b_0)x^2 + \cdots$

$h(x)$ → coefficient

$c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_{k-1} b_1 + a_k b_0$

→ Convolution of $f(x)$ & $g(x)$.

The seq c_k is a convolution of sequences a_k & b_k.

E.g. $f(x) = \frac{1}{(1-x)}$ $\quad g(x) = \frac{1}{1+x}$

$h(x) = f(x)g(x) = \frac{1}{1-x^2} = 1 + x^2 + x^4 + x^6 + \cdots$

The sequence $1, 0, 1, 0, \ldots$ is the convolution of $1, 1, 1, 1, \ldots$ and $1, -1, 1, -1, \ldots$

Partitions of Integers.

A partition of $n \in \mathbb{Z}^+$ writes n as a sum of positive parts where the order of the parts does not matter.

Claim: $n = 4$ has 5 partitions. $(p(n))$.

$p(4) : 5.$

① : $3+1$
② : $2+2$
③ : $1+1+1+1$
④ : $2+1+1$
⑤ : 4

→ Compute this for any general n. using GF method.

Obs1: A partition of n is completely determined by how many of each part is used.

Let $m_k \geq 0$ be the number of times part k is used.

The the partition of n is just a number of choices of $(m_1, m_2, m_3, \ldots)$.

$$1 \cdot m_1 + 2 \cdot m_2 + \ldots = n$$

Using part k, m_k times contributes $k m_k$

Record this as $\boxed{x^{k m_k}}$

$m_k = 0, 1, 2, \ldots$

All choices of part k can be collected by the GF: $1 + x^k + x^{2k} + \ldots = \frac{1}{1-x^k}$.

Vary this for all parts: $k \geq 1$.

$$\prod_{k \geq 1} \left(1 + x^k + x^{2k} + \ldots\right) = \prod_{k \geq 1} \frac{1}{1-x^k}.$$

In this GF $\frac{\text{coeff of } x^n}{}$ will tell me the no. of / count of partitions.

To get x^n → we can pick one term x^{km_k} from each factor of the above product

$$x^{1m_1} \; x^{2m_2} \cdots = x^n$$

$$= x^{m_1 + 2m_2 + \cdots} = x^n$$

Thus the coefficient of x^n in $\prod_{k \geq 1} \frac{1}{1-x^k}$ gives us the exact no. of partitions.

P(4) $k > 4$ is useless.

$$(1 + x + x^2 + x^3 + x^4)(1 + x^2 + x^4)(1 + x^3)(1 + x^4)$$

Coefficient of x^4.

$x^4 \cdot 1 \cdot 1 \cdot 1 = x^4$
$x^2 \cdot x^2 \cdot 1 \cdot 1 = x^4$
$x \cdot 1 \cdot x^3 \cdot 1 = x^4$
$1 \cdot x^4 \cdot 1 \cdot 1 = x^4$
$1 \cdot 1 \cdot 1 \cdot x^4 = x^4$

5.

E.g. Find the GF for the no. of ways an ad agent can purchase n minutes ($n \in \mathbb{Z}^+$) of air time if time slots for commercials come in blocks of 30, 60 & 120 seconds.

Hint: Take 30sec as the rep. of one unit time

$$a + 2b + 4c = 2n \qquad 0 \le a, b, c$$

$$f(x) = \left(1 + x + x^2 + \cdots\right)\left(1 + x^2 + x^4 + \cdots\right)\left(1 + x^4 + x^8 + \cdots\right)$$

$$= \frac{1}{1-x} \cdot \frac{1}{1-x^2} \cdot \frac{1}{1-x^4}$$

Soln → look for coeff of x^{2n} in the above expansion.

Counts the no. of ways in which $2n$ is partitioned into some 1's, some 2's & some 4's.

E.g. In how many ways can a police captain distribute 24 rifle shells to four police officers so that each officer gets at least three shells but not more than eight?

Use GF.

No. of choices a shell each officer receives is given by: $x^3 + x^4 + \cdots + x^8$

These are 4 officers:

$$(x^3 + x^4 + \cdots x^8)^4$$

Answer? Coeff of x^{24}.

$$x^{12}\left(1 + x^2 + \cdots + x^5\right)^4$$

$$x^{12}\left[\left(\frac{1-x^6}{1-x}\right)^4\right] \leftarrow \text{coeff of } x^{12}$$

$$\cdots \rightarrow \left[1 - \binom{4}{1}x^6 + \binom{4}{2}x^{12} - \binom{4}{3}x^{18} + x^{24}\right]\left[\binom{-4}{0} + \binom{-4}{1}(-x) + \binom{-4}{2}(-x)^2 + \cdots\right]$$

x^{12}.

$$\left[\binom{4}{12}(-1)^{12} - \binom{4}{1}\binom{-4}{6}(-1)^6 + \binom{4}{2}\binom{-4}{0}\right]$$

$$= \left[\binom{15}{12} - \binom{4}{1}\binom{9}{6} + \binom{4}{2}\right] = \boxed{125}$$

Defn: Exponential generating functions

Given a real sequence $a_0, a_1, a_2, \ldots$

$$f(x) = a_0 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \ldots$$

$$= \sum_{i \geq 0} a_i \frac{x^i}{i!}$$

If $a_i = 1$.

$$f(x) = 1 + x + \frac{x^2}{2!} + \ldots = e^x \text{ (Maclaurin series)}.$$

e^x is the exp. gf for the seq $[1\ 1\ 1\ 1\ 1 \ldots]$

e^x is the gf for the seq $[1, 1, 1/2!, 1/3! \ldots]$

Solve the recurrence relation:

$$a_n - 3a_{n-1} = n, \quad n \geq 1, \; a_0 = 1.$$

$n=1 \Rightarrow a_1 - 3a_0 = 1 \quad \ldots (1)$

$n=2 \Rightarrow a_2 - 3a_1 = 2 \quad \ldots (2)$

$\vdots$

Multiply (1) with x, (2) with x^2 . . .

$(n=1) \Rightarrow$

$$\begin{array}{|l|l|l|} a_1 x & - 3a_0 x & = 1x \\ a_2 x^2 & - 3a_1 x^2 & = 2x^2 \\ a_3 x^3 & - 3a_2 x^3 & = 3x^3 \\ \vdots & \vdots & \vdots \end{array}$$

$$\sum_{n\geq 1} a_n x^n - 3\sum_{n\geq 1} a_{n-1} x^n = \sum_{n\geq 1} n x^n$$

Call $f(x) = \sum_{n\geq 0} a_n x^n$ (GF).

$$(f(x) - a_0) - 3x\sum_{n\geq 0} a_{n-1}x^{n-1} = \sum_{n\geq 0} nx^n.$$

$$(f(x) - a_0) - 3xf(x) = \sum_{n\geq 0} nx^n.$$

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots$$

$$\frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + \cdots$$

$$(f(x) - a_0) - 3xf(x) = \frac{x}{(1-x)^2}$$

$$f(x) - 1 - 3xf(x) = \frac{x}{(1-x)^2}$$

$$f(x) = \frac{1}{(1-3x)} + \frac{x}{(1-x)^2(1-3x)}.$$

partial fraction decomposition

$$\frac{x}{(1-x)^2(1-3x)} = \frac{A}{(1-x)} + \frac{B}{(1-x)^2} + \frac{C}{(1-3x)} \Rightarrow B = -1/2,\ C = 3/4,$$
$$A = -1/4$$

$$f(x) = \frac{7/4}{(1-3x)} + \frac{-1/4}{(1-x)} + \frac{-1/2}{(1-x)^2}$$

Determine a_n → coefficient of x^n

(a) $\left(\frac{7}{4}\right)/(1-3x) = \frac{7}{4}\left[1 + 3x + (3x)^2 + \dots\right]$

Co. eff of $x^n = \frac{7}{4} \cdot 3^n$.

(b) $\left(-\frac{1}{4}\right)/(1+x) = -\frac{1}{4}\left[1 + x + x^2 + \dots\right]$.

Co. eff of $x^n = -\frac{1}{4} 1^n = -\frac{1}{4}$

(c) $-1/2/(1-x)^2 = \left(-\frac{1}{2}\right)(1-x)^{-2}$

$= -\frac{1}{2}\left[\binom{-2}{0} + \binom{-2}{1}(-x) + \dots\right]$

Co. eff $x^n \rightarrow -\frac{1}{2} \cdot \binom{-2}{n} \cdot (-1)^n$

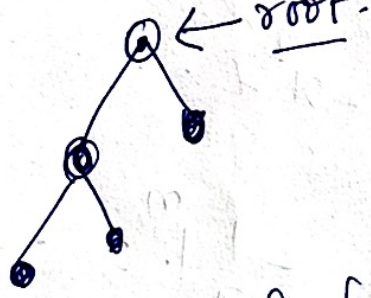
$= -\frac{1}{2} \cdot (-1)^n \binom{2+n-1}{n} (-1)^n = -\frac{1}{2}(n+1)$

Full sol^{n} = (a) + (b) + (c)

$$a_n = \left(\frac{7}{4}\right)3^n + \frac{n}{2} - \frac{3}{4}, \quad n \geq 0.$$

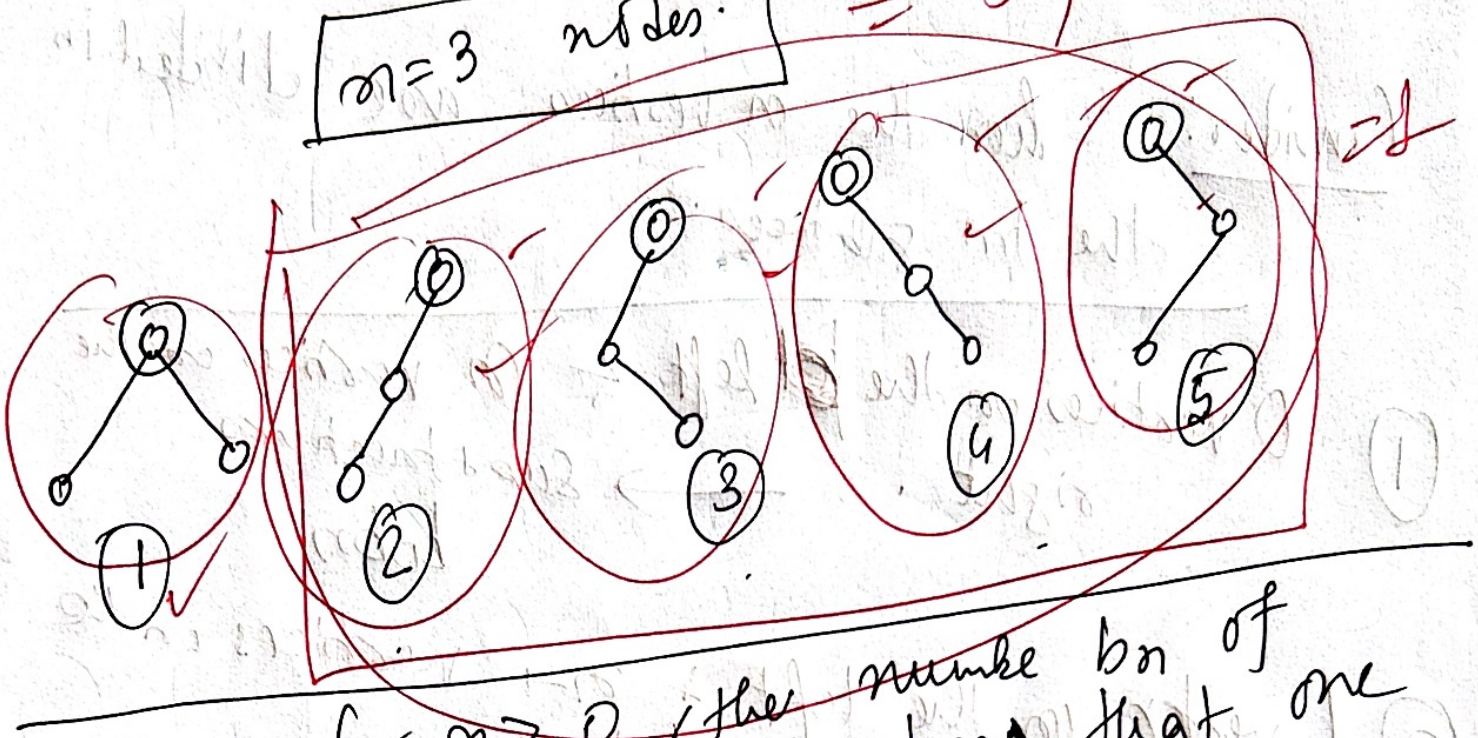
✗ No need for $a_n^{(h)}$ & $a_n^{(p)}$ separately

Rooted binary trees.



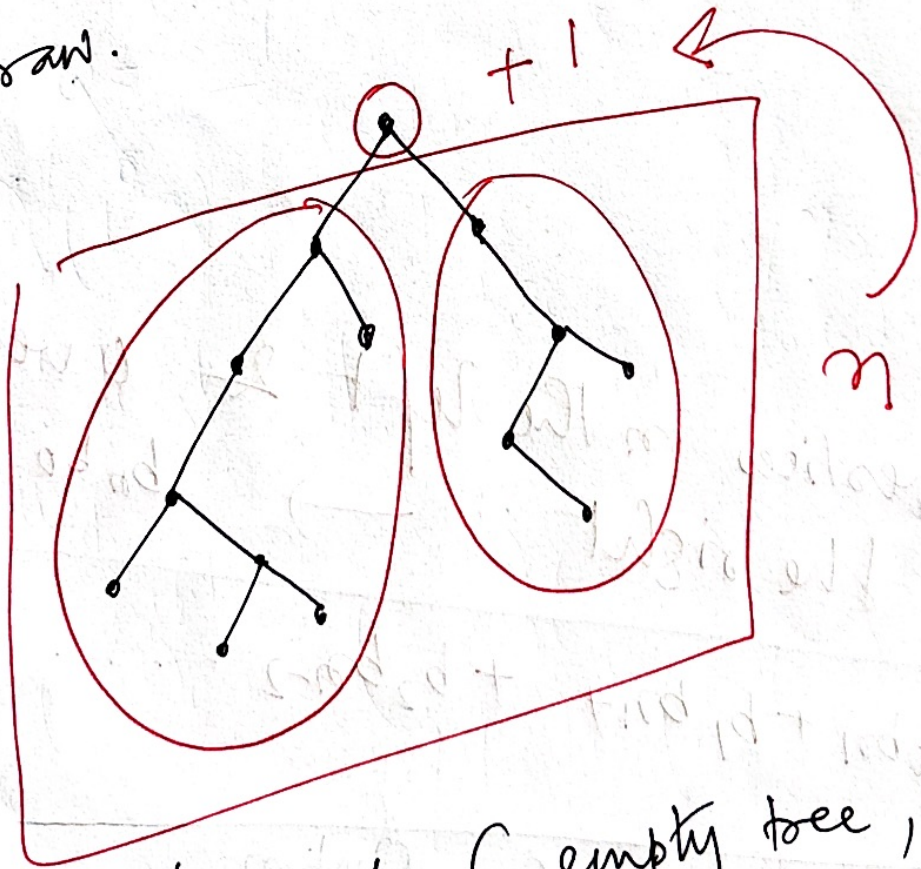
Rooted ordered binary tree: Left branch descending from a vertex is considered different from the right branch.

$n = 3$ nodes.



Count for $n \geq 0$ (the number of rooted ordered binary trees that one can draw.

$= n+1$.



$b_0 = 1$ (empty tree, can be only 1).

Consider: How the n vertices are divided in the two subtrees.

(1) 0 vertices on the left & n vertices on the right. → Substructures $b_0 b_n$

(2) 1 vertex on the left & $n-1$ vertices on the right → $b_1 b_{n-1}$

⋮

$(n+1)$ n vertices on the left & 0 vertices on the right → $b_n b_0$ substructures

$$b_{n+1} = b_0 b_n + b_1 b_{n-1} + b_2 b_{n-2} + \cdots + b_n b_0$$

Convolution.

$$\sum_{n\geq 0} b_{n+1}x^{n+1} = \sum_{n\geq 0}\left(b_0 b_n + b_1 b_{n-1} + \cdots + b_n b_0\right)x^{n+1}$$

$$f(x) = \sum_{n=0}^{\infty} b_n x^n.$$

$$f(x) - b_0 = x\sum_{n\geq 0}\left(b_0 b_n + b_1 b_{n-1} + \cdots + b_n b_0\right)x^n$$

$$= x[f(x)]^2$$

$$x[f(x)]^2 - f(x) + 1 = 0.$$

$$f(x) = \frac{1 \pm \sqrt{1-4x}}{2x}.$$

$$\sqrt{1-4x} = \sum_{k\geq 0}\binom{1/2}{k}(-4x)^k.$$

x^k coefficient $\Rightarrow \binom{1/2}{k}(-4)^k = -\frac{2}{k}\binom{2k-2}{k-1}$.

$$\frac{1 \pm \sqrt{1-4x}}{2x} = \frac{1 - 2\sum_{k\geq 1} \frac{1}{k}\binom{2k-2}{k-1} x^k}{2x}$$

$$\frac{1-\sqrt{1-4x}}{2x} = \frac{1 - \left[1 - 2\sum_{k\geq 1} \frac{1}{k}\binom{2k-2}{k-1} x^k\right]}{2x}$$

$$= \sum_{k\geq 1} \frac{1}{k}\binom{2k-2}{k-1} x^{k-1}$$

$$k = n+1$$

$b_{n+1} \Rightarrow$

coefficient $k = n+1$ will be.

$$\frac{1}{n+1}\binom{2n}{n} \leftarrow \text{Catalan numbers.}$$